

Glow Logic – Architecture fonctionnelle Frontend

Aperçu

Ce document décrit l'architecture fonctionnelle du frontend de Glow Logic autour de 6 modules interconnectés. L'objectif est de garantir une UI cohérente, une synchronisation en temps réel via un store Zustand unique et des flux de données clairement définis pour DMX, MIDI, audio et 3D.

1. Orchestrateur de l'interface (Layout & Zustand Store)

Store Central (useStore.ts)

- Slices: appMode (smart | live | creator), proView (canvas | patch | visualizer), groups[{id,name, intensity,colorHSV,muted,solo}], palettesActives, master, blackout, strobe, effets, selection{fixtures,groups}, patch{univers,fixtures,profiles}, mediaState, networkState, midiState.
- Actions: setAppMode, setProView, setGroupIntensity/Color/Muted/Solo, setMaster/Blackout/Strobe, selectFixture/Group, setPatch/Profiles, setMediaState, setNetworkState, setMidiMapping, setAudioLevels.
- Selectors mémorisés par vue; persistance (zustand/persist). Composants fixes
- TopBar: ModeSwitch, ProViewNav, Net/MIDI indicators, QuickMaster.
- SidebarLeft/Right: panneaux contextuels dynamiques selon mode (smart/live/creator).
- MacroTimeline: séquenceur bas d'écran, play/stop/record, bind au store (tempo, cues).

2. Module Smart Dashboard (Console de Régie Modulaire)

Composants

- DmxMixer6: 6 faders verticaux, read/write store.groups[i].intensity; ColorPickerHSV par groupe; Mute/Solo; envoi d'événements DMX via adaptateur.
- ScenePadsGrid: layout responsive 2-8 colonnes, Drag&Drop (dnd-kit), PadConfigModal (nom, palette RGBW, icône, mapping MIDI canal/note). Pads déclenchent des scènes → actions sur store (intensités/couleurs/effets).
- ApcMiniVirtual: grille 8x8, feedback LED simulé; binding à MidiListener/ApcMiniMapper et à ScenePadsGrid.
- AudioAnalyzer + SafePlaylist: FFT bas/mid/high, thresholds dans store.setAudioLevels; moteur d'auto-triggers vers scènes; lecteur playlist de secours.

3. Module Live Performance View (Régie Spectacle)

Composants

- MediaBrowserSidebar: arborescence, preview, métadonnées → updates mediaState.
- LiveCenter: ApcMiniVirtual superposé au PlanDeFeu2DView; affiche états intensity/color/selection en temps réel.

- QuickControls: Master, Blackout, Strobe, Effets; écritures directes store.

4. Module Creator & Configuration

Composants

- PlanDeFeu2D: grille orthonormée, drag&drop fixtures, align/distribute, groupes visuels, import/export JSON; met à jour patch et selection.
- DmxSetupWizard: scan interfaces réseau, IP fixes, Art-Net nodes, USB FTDI/Enttec; tests; commit vers store.patch/networkState.
- FixtureProfileBuilder: éditeur (fabricant, canaux, modes); IA parser de texte PDF collé → Footprint; push vers store.profiles.

5. Module Visualiseur 3D & Mini-Overlay

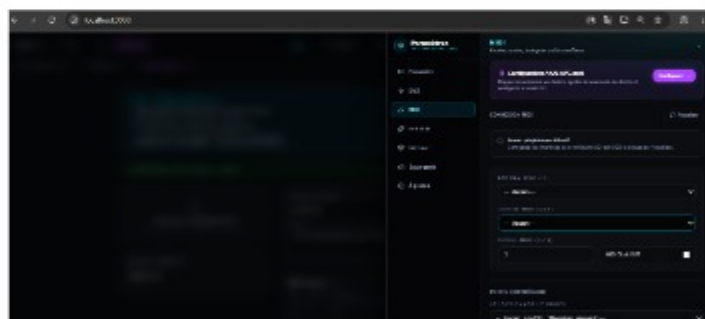
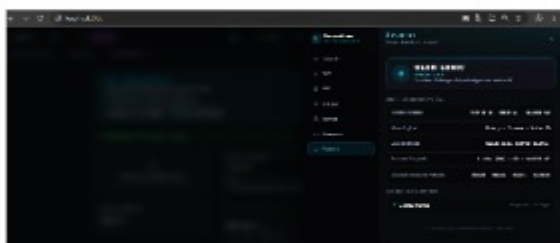
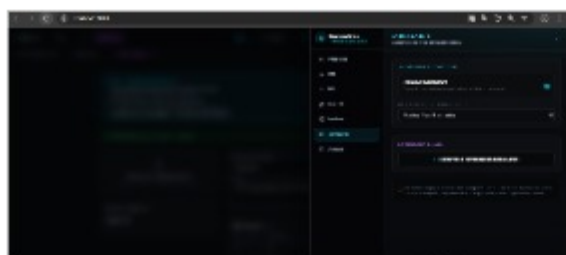
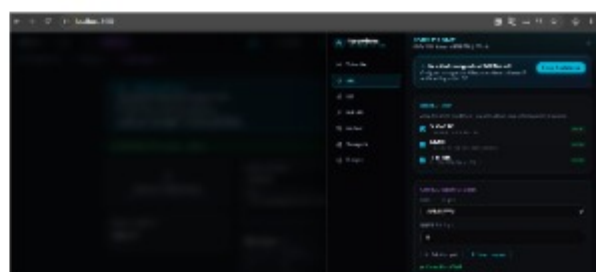
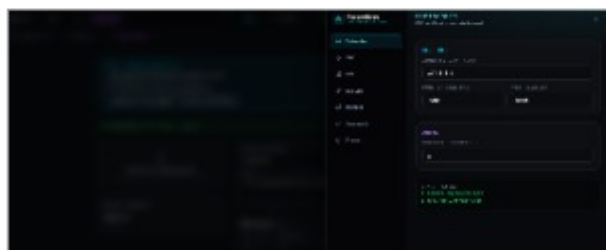
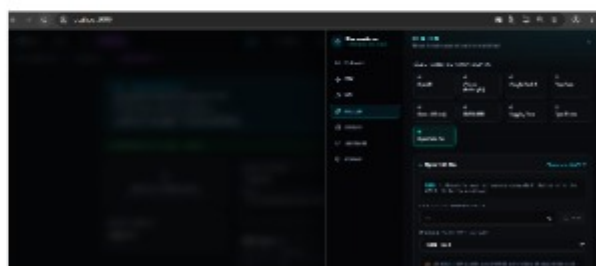
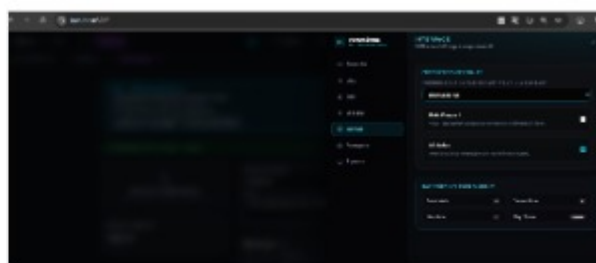
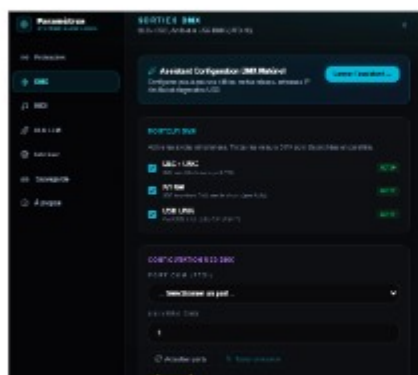
Composants

- Visualizer3D (Three.js): lyres 7R, pan/tilt, gobo/prisme/zoom, volumétrie; lit patch/values du store; sélection par clic → store.selection.
- MiniOverlay.tsx: fenêtre flottante S/M/L, toggle 2D/3D, draggable, minimisable; quick controls intensité/couleur par groupe.

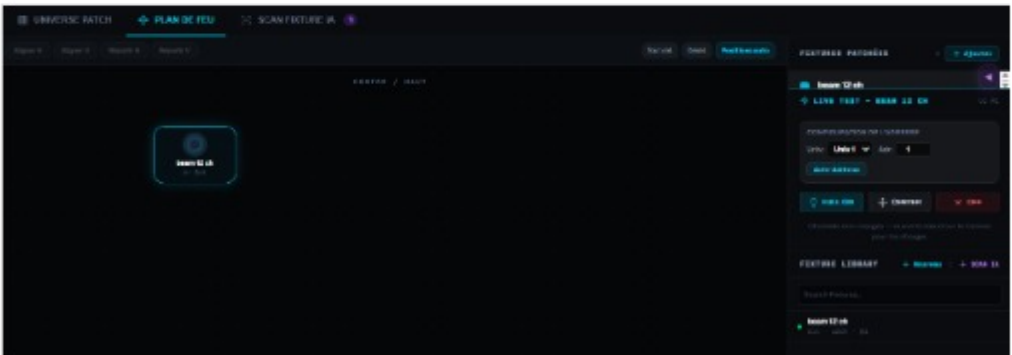
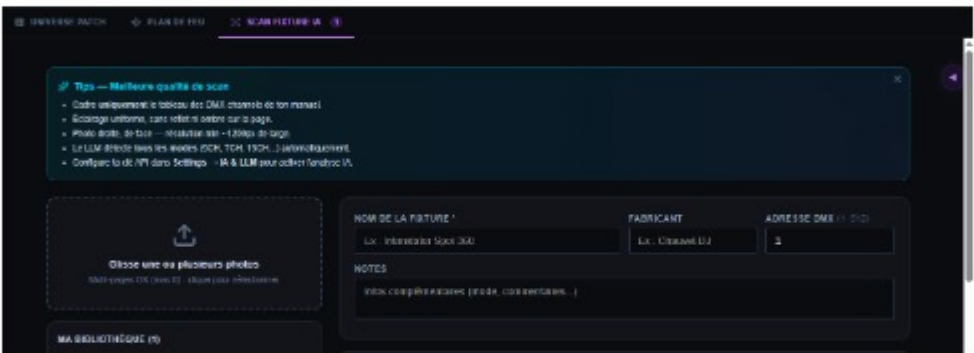
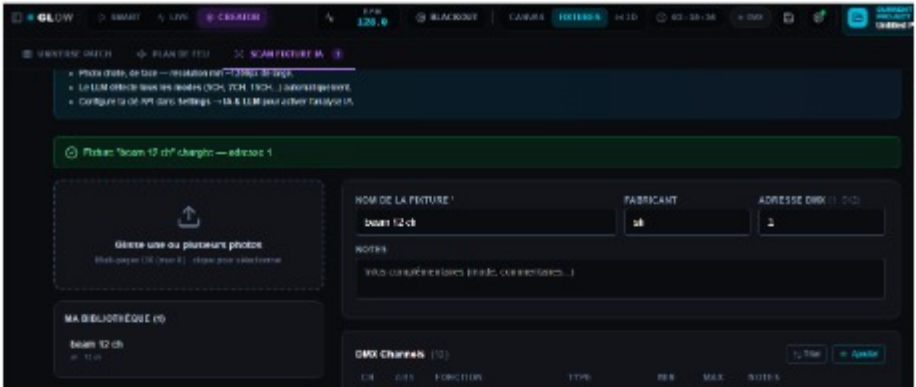
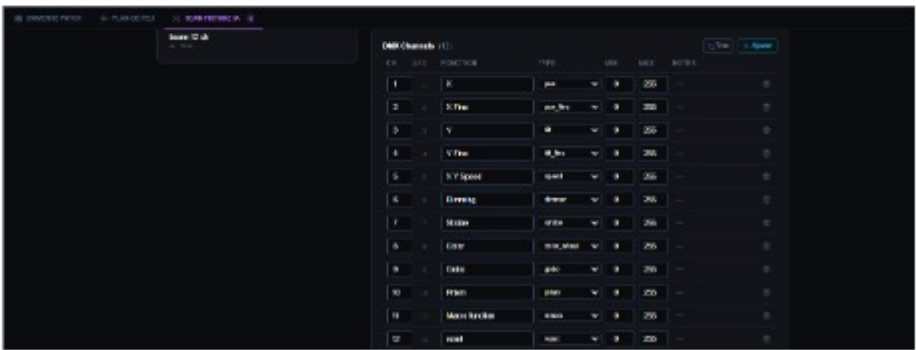
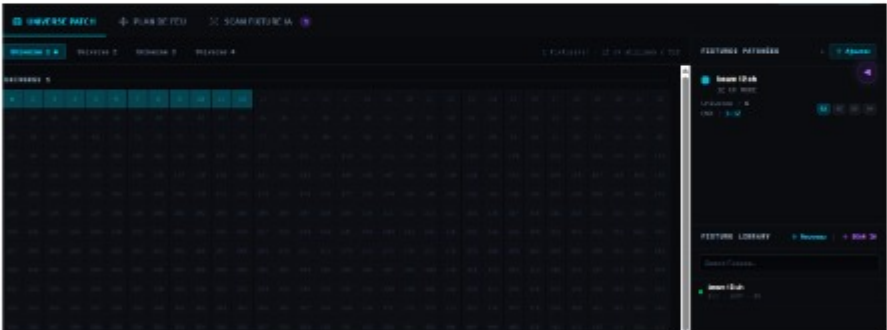
6. Flux et Connexions

- useStore ↔ SmartDashboard, LivePerformance, Creator, Visualizer3D, Overlays (synchronisation UI/valeurs).
- PlanDeFeu2D ↔ Visualizer3D: positions/orientations et sélection bidirectionnelles.
- MidiListener/ApcMiniMapper → store → ApcMiniVirtual/ScenePadsGrid.
- DmxSetupWizard → store (réseau/univers) → adaptateurs DMX.
- FixtureProfileBuilder → store.profiles → PlanDeFeu2D/Patch.
- AudioAnalyzer → store.audio → déclenchement de scènes.

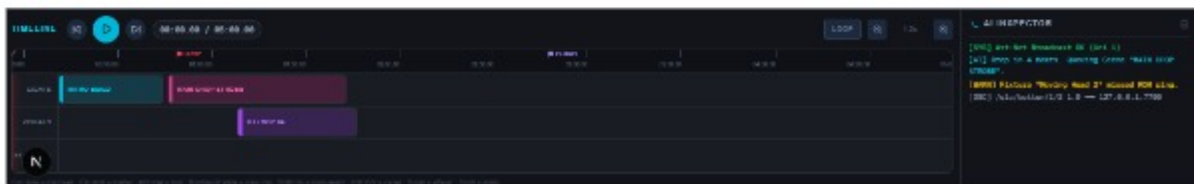
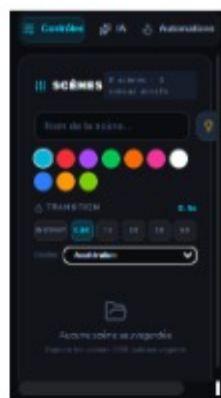
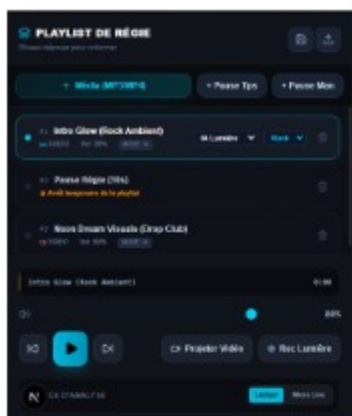
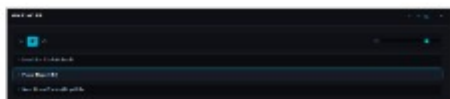
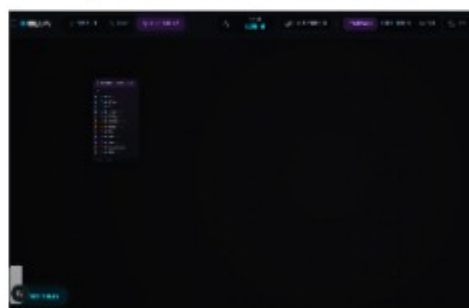
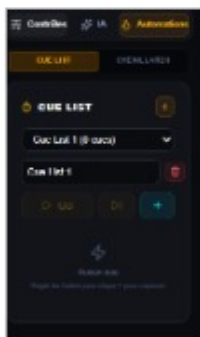
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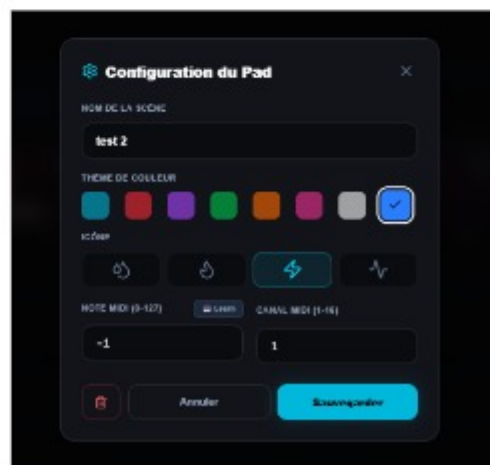
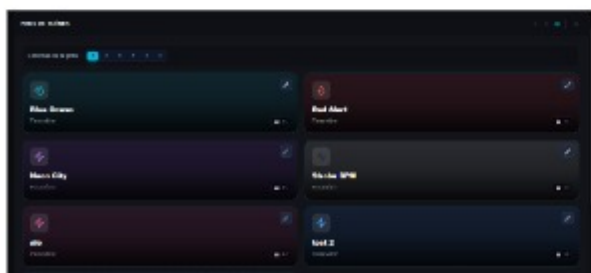
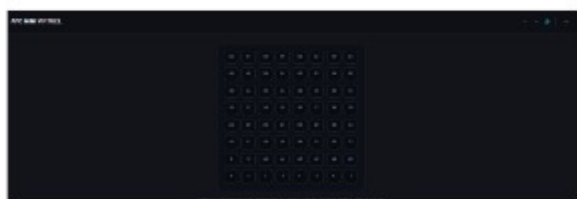
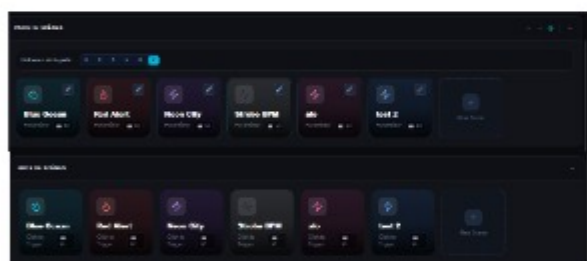
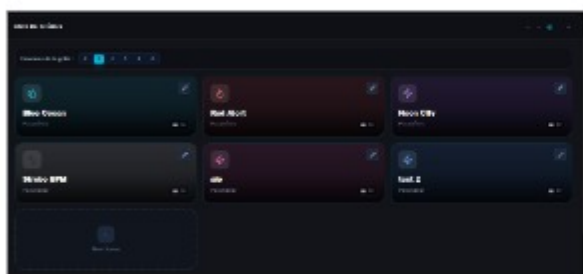
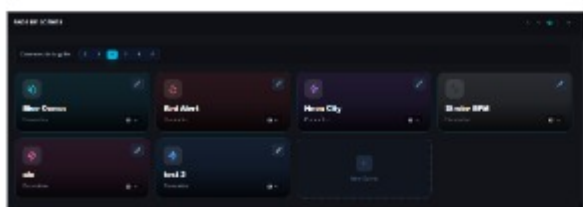
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patch scan
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line ,edition
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,arrangement



live,control
,mapping
,assignation



sidebar
an
other

